

DEPARTMENT OF MATHEMATICS · BS-IV

Statistical Inference

Complete Workbook

$\Sigma \hat{\theta} \chi^2 \mathcal{L}$

Probability · Estimation · Testing · Regression · Bayesian

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BS-IV Mathematics · 2026

Chapters at a Glance

Ch. 1. Introduction to Statistics
Ch. 2. Probability — Foundations
Ch. 3. Probability — Card Games
Ch. 4. Descriptive Statistics
Ch. 5. Sampling Distributions
Ch. 6. Point Estimation

Ch. 7. Interval Estimation
Ch. 8. Hypothesis Testing
Ch. 9. Regression & Correlation
Ch. 10. Non-Parametric Tests
Ch. 11. Bayesian Inference

Each chapter: historical background · formal proofs · full derivations · worked examples · practice problems.

Scope: A rigorous BS-IV treatment of Statistical Inference. Every major result is stated with complete proof or derivation, following a logical progression: probability foundations, descriptive statistics, sampling theory, estimation, hypothesis testing, regression, non-parametric methods, and Bayesian inference. The card-game chapter builds probabilistic intuition in an applied, visual setting.

Table of Contents

Chapter 1 — Introduction to Statistics	3
Chapter 2 — Probability: Foundations	6
Chapter 3 — Probability: Card Games	12
Chapter 4 — Descriptive Statistics	16
Chapter 5 — Sampling Distributions	20
Chapter 6 — Point Estimation	24
Chapter 7 — Interval Estimation	29
Chapter 8 — Hypothesis Testing	32
Chapter 9 — Regression & Correlation	36
Chapter 10 — Non-Parametric Tests	40
Chapter 11 — Bayesian Inference	43
Quick Reference	46

CHAPTER 01

Introduction to Statistics

Historical Background

The word *statistics* derives from the Latin *statisticum collegium* (council of state) and the Italian *statista* (statesman). Its earliest use was the collection of data about the state — population, taxation, and military resources. **John Graunt** (1662) analysed London's Bills of Mortality and is credited as the founder of demography. **Karl Pearson** (1857–1936) and **Ronald A. Fisher** (1890–1962) transformed statistics into a formal mathematical discipline in the early 20th century. Fisher introduced the concepts of likelihood, sufficiency, and experimental design. **Jerzy Neyman** and **Egon Pearson** developed the modern theory of hypothesis testing in the 1930s. Today, statistics is indispensable in every scientific field.

Definition — Statistics

Statistics is the science of collecting, organising, summarising, analysing, and interpreting numerical data in order to draw valid conclusions and make decisions under uncertainty.

1.1 Descriptive vs Inferential Statistics

Definition — Two Branches

Descriptive Statistics summarises and presents data in a meaningful form using tables, charts, measures of centre, and measures of spread. It describes *what the data shows* — no generalisation beyond the data.

Inferential Statistics uses sample data to draw conclusions about a larger population. It involves estimation, hypothesis testing, and prediction — always with a measure of uncertainty.

Example 1.1 — Descriptive vs Inferential

A class of 40 students is given a maths test.

Descriptive: “The average score of these 40 students is 72.” (fact about the sample)

Inferential: “Based on this sample, we estimate that the average score of all students in the school is between 68 and 76 with 95% confidence.” (generalisation to the population)

1.2 Population, Sample, Parameter, Statistic

Definition — Key Terms

- **Population:** the complete collection of all individuals, items, or measurements of interest
- **Sample:** a subset of the population, selected for study
- **Parameter:** a fixed numerical characteristic of the population (Greek letters: μ , σ^2 , p). Usually *unknown*.
- **Statistic:** a numerical summary computed from sample data (Roman letters: $\bar{x}$, s^2 , $\hat{p}$). Used to *estimate* parameters.

Measure	Population Symbol	Sample Symbol	Formula (Sample)
Mean	μ	$\bar{x}$	$\frac{1}{n} \sum_{i=1}^n x_i$
Variance	σ^2	s^2	$\frac{\sum (x_i - \bar{x})^2}{n-1}$
Std Dev	σ	s	$\sqrt{s^2}$
Proportion	p	$\hat{p}$	x/n
Size	N	n	—

1.3 Types of Data and Measurement Scales

Definition — Types of Variables

Qualitative (Categorical): names or categories; no numerical meaning. *E.g. gender, colour, religion.*

Quantitative — Discrete: countable values. *E.g. number of children, number of defects.*

Quantitative — Continuous: any value in an interval. *E.g. height, weight, temperature.*

Definition — Measurement Scales (Stevens, 1946)

1. **Nominal:** categories with no order. *E.g. blood group (A, B, AB, O)*
2. **Ordinal:** ordered categories but differences are meaningless. *E.g. rank (1st, 2nd, 3rd)*
3. **Interval:** ordered with meaningful differences but no true zero. *E.g. temperature in °C*
4. **Ratio:** interval scale with a true zero. *E.g. height, weight, income*

The scale determines which statistical operations are valid.

1.4 Sampling Methods

Definition — Types of Probability Sampling

1. **Simple Random Sampling (SRS):** every sample of size n is equally likely. Best for homogeneous populations.
2. **Stratified Sampling:** population divided into non-overlapping strata; SRS from each stratum. Better representation of subgroups.
3. **Systematic Sampling:** list population; select every k -th unit after random start. Simple to implement.
4. **Cluster Sampling:** population divided into clusters; randomly select whole clusters. Economical for geographically dispersed populations.

CHAPTER 02

Probability — Foundations

Historical Background

The formal mathematical theory of probability began with a correspondence between **Blaise Pascal** and **Pierre de Fermat** in 1654, sparked by the “Problem of Points” — how to divide stakes when a game of chance is interrupted. **Christiaan Huygens** published the first treatise on probability in 1657. **Jakob Bernoulli**’s *Ars Conjectandi* (1713) proved the Law of Large Numbers. **Abraham de Moivre** discovered the normal distribution (1733). **Pierre-Simon Laplace** unified and extended the theory in *Théorie Analytique des Probabilités* (1812). The modern axiomatic foundation was established by **Andrey Kolmogorov** in 1933.

Definition — Probability — Three Definitions

Classical (Laplace): If N outcomes are equally likely and A occurs in m of them:

$$P(A) = \frac{m}{N} = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

Empirical (Frequency): $P(A) = \lim_{n \rightarrow \infty} \frac{f_A}{n}$ where f_A is the frequency of A in n trials.

Axiomatic (Kolmogorov, 1933): P is a function on events satisfying three axioms (see below).

2.1 Sample Space and Events

Definition — Sample Space, Events, Set Operations

Sample space S : the set of all possible outcomes.

Event $A \subseteq S$: any subset of S .

Set operations:

- **Union $A \cup B$:** outcomes in A or B (or both)
- **Intersection $A \cap B$:** outcomes in both A and B
- **Complement A^c :** outcomes not in A ; $P(A^c) = 1 - P(A)$
- **Difference $A \setminus B = A \cap B^c$:** in A but not B
- **Mutually exclusive:** $A \cap B = \emptyset$ (cannot both occur)
- **Exhaustive:** $A_1 \cup \dots \cup A_k = S$

2.2 Kolmogorov's Axioms

Theorem — Kolmogorov Axioms (1933)

A probability measure P on sample space S satisfies:

- A1.** $P(A) \geq 0$ for every event A (non-negativity)
- A2.** $P(S) = 1$ (normalisation)
- A3.** If $A_1, A_2, \dots$ are mutually exclusive: $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ (countable additivity)

All other probability laws are **derived** from these three axioms.

Derivation — Derived Rules from Axioms

- (i) $P(\emptyset) = 0$: S and $\emptyset$ are mutually exclusive with $S \cup \emptyset = S$. By A3: $P(S) + P(\emptyset) = P(S) = 1$, so $P(\emptyset) = 0$.
- (ii) $P(A^c) = 1 - P(A)$: A and A^c are mutually exclusive with $A \cup A^c = S$. By A3: $P(A) + P(A^c) = 1$.
- (iii) $0 \leq P(A) \leq 1$: From A1, $P(A) \geq 0$. Since $A \subseteq S$, by monotonicity $P(A) \leq P(S) = 1$.
- (iv) **If $A \subseteq B$ then $P(A) \leq P(B)$** : Write $B = A \cup (B \setminus A)$ disjointly. Then $P(B) = P(A) + P(B \setminus A) \geq P(A)$.

2.3 Simple and Compound Events

Definition — Simple vs Compound Events

A **simple (elementary) event** consists of a single outcome: $A = \{\omega\}$ for some $\omega \in S$.

A **compound event** consists of two or more outcomes, formed by combining simple events.

Example: Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$. “Getting 4” is simple; “getting an even number” $= \{2, 4, 6\}$ is compound.

2.4 Addition Theorem

Theorem — Addition Theorem

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof: Decompose $A \cup B$ into three disjoint pieces: $A \cup B = (A \cap B^c) \cup (A \cap B) \cup (A^c \cap B)$

By A3: $P(A \cup B) = P(A \cap B^c) + P(A \cap B) + P(A^c \cap B)$

Also: $P(A) = P(A \cap B^c) + P(A \cap B)$ and $P(B) = P(A^c \cap B) + P(A \cap B)$.

Adding: $P(A) + P(B) = P(A \cap B^c) + P(A^c \cap B) + 2P(A \cap B) = P(A \cup B) + P(A \cap B)$.

Rearranging: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $\square$

For three events: $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

Neither A nor B means the complement of their union:

$$P(\text{neither } A \text{ nor } B) = 1 - P(A \cup B) = 1 - P(A) - P(B) + P(A \cap B)$$

Example 2.1 — Addition Theorem

In a group of 200 students: 80 study Maths (M), 70 study Physics (P), 30 study both.

(a) $P(M \cup P) = 80/200 + 70/200 - 30/200 = 120/200 = 0.60$

(b) $P(\text{neither}) = 1 - 0.60 = 0.40$

(c) $P(M \text{ only}) = P(M) - P(M \cap P) = 80/200 - 30/200 = 50/200 = 0.25$

2.5 Conditional Probability and Multiplication Theorem

Definition — Conditional Probability

The probability of A given B has occurred:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Intuition: we restrict the sample space from S to B , then ask how much of B is also in A .

Theorem — Multiplication Theorem

From the definition of conditional probability:

$$P(A \cap B) = P(A | B) \cdot P(B) = P(B | A) \cdot P(A)$$

For three events: $P(A \cap B \cap C) = P(A) \cdot P(B | A) \cdot P(C | A \cap B)$.

Example 2.2 — Conditional Probability

A box has 5 red and 3 blue balls. Two are drawn without replacement.

$$P(2\text{nd red} \mid 1\text{st red}) = 4/7.$$

$$P(\text{both red}) = P(1\text{st red}) \times P(2\text{nd red} \mid 1\text{st red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14} \approx 0.357$$

2.6 Statistical Independence**Definition — Independent Events**

A and B are **statistically independent** if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Equivalently: $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$ — knowledge of one event gives no information about the other.

Important: Mutually exclusive events (with $P(A), P(B) > 0$) are *never* independent. If $A \cap B = \emptyset$ then $P(A \cap B) = 0 \neq P(A)P(B)$.

2.7 Total Probability Theorem**Theorem — Law of Total Probability**

If $B_1, B_2, \dots, B_k$ partition S (mutually exclusive and exhaustive, each with $P(B_i) > 0$), then for any event A :

$$P(A) = \sum_{i=1}^k P(A \mid B_i) P(B_i)$$

Proof: Since $\{B_i\}$ partition S : $A = A \cap S = A \cap \bigcup_i B_i = \bigcup_i (A \cap B_i)$.

The events $A \cap B_i$ are mutually exclusive, so by A3: $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A \mid B_i) P(B_i)$. $\square$

2.8 Bayes' Theorem**Theorem — Bayes' Theorem (1763)**

Under the same partition $\{B_1, \dots, B_k\}$:

$$P(B_i \mid A) = \frac{P(A \mid B_i) P(B_i)}{\sum_{j=1}^k P(A \mid B_j) P(B_j)}$$

Proof: By conditional probability: $P(B_i \mid A) = P(A \cap B_i) / P(A) = P(A \mid B_i) P(B_i) / P(A)$. Substitute the total probability formula for $P(A)$. $\square$

Terminology: $P(B_i) = \text{prior probability}$; $P(B_i \mid A) = \text{posterior probability}$.

Example 2.3 — Bayes' Theorem: Disease Testing

A disease affects 1% of the population. A test has: sensitivity (true positive rate) = 95%; specificity (true negative rate) = 90%. A patient tests positive. What is the probability they have the disease?

Let D = has disease, T^+ = tests positive.

$$P(D) = 0.01, P(D^c) = 0.99, P(T^+ | D) = 0.95, P(T^+ | D^c) = 0.10.$$

$$P(T^+) = 0.95(0.01) + 0.10(0.99) = 0.0095 + 0.099 = 0.1085$$

$$P(D | T^+) = \frac{0.95 \times 0.01}{0.1085} = \frac{0.0095}{0.1085} \approx \mathbf{0.0876} \approx 8.8\%$$

Interpretation: Even with a positive test, there is only $\approx 8.8\%$ chance of having the disease due to the low prevalence. This is the **base rate fallacy** — a critical concept in medical statistics.

2.9 Counting Principles**Definition — Fundamental Counting Rules**

Multiplication Rule: If task 1 can be done in m ways and task 2 in n ways, both together: $m \times n$ ways.

Permutation (order matters):

$${}_nP_r = P(n, r) = \frac{n!}{(n-r)!}$$

Combination (order does not matter):

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Key identity: $\binom{n}{r} = \binom{n}{n-r}$; $\binom{n}{0} = \binom{n}{n} = 1$.

Example 2.4 — Counting

(a) 8 runners in a race. How many ways can 1st, 2nd, 3rd place be awarded?

$$P(8, 3) = 8 \times 7 \times 6 = 336$$

(b) A committee of 4 is chosen from 10 people. How many ways?

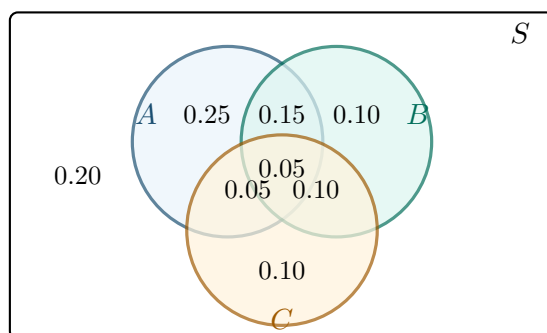
$$\binom{10}{4} = 210$$

(c) From 6 men and 4 women, select 3 men and 2 women: $\binom{6}{3} \times \binom{4}{2} = 20 \times 6 = 120$

2.10 Venn Diagrams

Example 2.5 — Venn Diagram with Three Events

In a class: $P(A) = 0.5$, $P(B) = 0.4$, $P(C) = 0.3$, $P(A \cap B) = 0.2$, $P(A \cap C) = 0.1$, $P(B \cap C) = 0.15$, $P(A \cap B \cap C) = 0.05$.



$$P(A \cup B \cup C) = 0.5 + 0.4 + 0.3 - 0.2 - 0.1 - 0.15 + 0.05 = 0.80. \quad P(\text{none}) = 0.20.$$

CHAPTER 03

Probability — Card Games

Historical Background

Playing cards arrived in Europe from Asia around the 14th century. The standard 52-card deck evolved in France around 1480, where the four suits — spades (♠), hearts (♥), diamonds (♦), clubs (♣) — became standardised. Mathematicians began analysing card games as early as the 17th century; Gerolamo Cardano's *Liber de Ludo Aleae* (c. 1560) computed probabilities for dice and cards. The analysis of poker hands was a landmark early application of combinatorial probability. Today, card-based probability problems appear in every undergraduate statistics course as a rich, visual setting to develop probabilistic thinking.

3.1 Structure of the Standard Deck

♥ Hearts	♦ Diamonds	♠ Spades	♣ Clubs
13 cards	13 cards	13 cards	13 cards
Red: 26 cards		Black: 26 cards	

Category	Count	Cards
Aces	4	A♥ A♦ A♠ A♣
Kings	4	K♥ K♦ K♠ K♣
Queens	4	Q♥ Q♦ Q♠ Q♣
Jacks	4	J♥ J♦ J♠ J♣
Face Cards	12	J, Q, K in each of 4 suits
Number Cards	36	2–10 in each of 4 suits

Sample 5-card hand:



3.2 Single-Card Probabilities

Example 3.1 — Basic Single-Card Draws

One card drawn at random from a shuffled 52-card deck:

Event	Count	Probability
Ace	4	$4/52 = 1/13 \approx 0.077$
Heart (♥)	13	$13/52 = 1/4 = 0.25$
Face Card (J,Q,K)	12	$12/52 = 3/13 \approx 0.231$
Red Card	26	$26/52 = 1/2 = 0.50$
Black Ace	2	$2/52 = 1/26 \approx 0.038$
Red Face Card	6	$6/52 = 3/26 \approx 0.115$
Not a King	48	$48/52 = 12/13 \approx 0.923$

3.3 Compound Events and Neither/Nor

Example 3.2 — Union, Intersection, Neither/Nor

One card drawn. Let A = Queen, B = Spade.

$P(A) = 4/52$, $P(B) = 13/52$, $P(A \cap B) = P(\text{Queen of Spades}) = 1/52$.

(a) Queen or Spade:

$$P(A \cup B) = \frac{4 + 13 - 1}{52} = \frac{16}{52} = \frac{4}{13} \approx 0.308$$

(b) Neither Queen nor Spade:

$$P((A \cup B)^c) = 1 - \frac{16}{52} = \frac{36}{52} = \frac{9}{13} \approx 0.692$$

(c) Queen but not Spade: $P(A \cap B^c) = 4/52 - 1/52 = 3/52$

These are:   

Example 3.3 — Conditional Probability with Cards

Given the drawn card is a **Face Card**, find the probability it is a King.

Face cards (12 total):            

Kings among face cards: 4.

$$P(\text{King} \mid \text{Face}) = \frac{P(\text{King} \cap \text{Face})}{P(\text{Face})} = \frac{4/52}{12/52} = \frac{4}{12} = \frac{1}{3} \approx 0.333$$

3.4 Drawing Multiple Cards

Definition — With vs Without Replacement

With replacement: after drawing, the card is returned. Each draw is **independent**: $P(\text{2nd event})$ unchanged.

Without replacement: drawn card is *not* returned. Draws are **dependent**: each draw reduces the deck by one.

Example 3.4 — Two-Card Draws: With and Without Replacement

Find $P(\text{both Aces})$:

With replacement: $P = \frac{4}{52} \times \frac{4}{52} = \frac{16}{2704} = \frac{1}{169} \approx 0.00592$

Without replacement: $P = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221} \approx 0.00452$

The without-replacement probability is lower because one Ace has been removed from the deck.

Find $P(\text{first Ace, second King})$ without replacement:

$$P = \frac{4}{52} \times \frac{4}{51} = \frac{16}{2652} = \frac{4}{663} \approx 0.00603$$

Example 3.5 — Three Cards Without Replacement

Three cards drawn without replacement. Find $P(\text{all three are Hearts})$.

$$P = \frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} = \frac{1716}{132600} = \frac{11}{850} \approx 0.01294$$

3.5 Five-Card Poker Hands

Definition — Total 5-Card Hands

$$\binom{52}{5} = \frac{52!}{5!47!} = 2\,598\,960$$

Example 3.6 — Poker Hand Probabilities

(a) **Royal Flush** (A, K, Q, J, 10 of same suit): 4 ways (one per suit).

$$P = \frac{4}{2\,598\,960} \approx 0.00000154$$



(b) **Four of a Kind:** choose rank $\binom{13}{1}$, all 4 suits, plus any 1 of 48 remaining cards.

$$\binom{13}{1} \times 1 \times 48 = 624 \quad P \approx 0.000240$$



(c) **Full House:** choose rank for triple $\binom{13}{1}$, choose 3 suits $\binom{4}{3}$; choose rank for pair $\binom{12}{1}$, choose 2 suits $\binom{4}{2}$.

$$13 \times 4 \times 12 \times 6 = 3744 \quad P \approx 0.00144$$



(d) **Flush** (5 same suit, not straight): $4 \times \binom{13}{5} - 40 = 5108$. $P \approx 0.00197$.

(e) **One Pair:** Choose rank $\binom{13}{1}$, suits $\binom{4}{2}$; choose 3 different ranks $\binom{12}{3}$, 1 suit each 4^3 :

$$13 \times 6 \times 220 \times 64 = 1\,098\,240 \quad P \approx 0.4226$$

Hand	Combinations	Probability
Royal Flush	4	0.00000154
Straight Flush	36	0.0000139
Four of a Kind	624	0.000240
Full House	3,744	0.00144
Flush	5,108	0.00197
Straight	10,200	0.00392
Three of a Kind	54,912	0.0211
Two Pair	123,552	0.0475
One Pair	1,098,240	0.4226
High Card	1,302,540	0.5012
Total	2,598,960	1.0000

Practice Problems

- Q1. From 52 cards, find $P(\text{Ace or Club})$.
 Q2. Three cards drawn without replacement. Find $P(\text{first two red, third black})$.
 Q3. A card is a red card. Given this, find $P(\text{it is an Ace})$.
 Q4. Find $P(\text{neither a heart nor a face card})$ when one card is drawn.
 Q5. Two cards drawn without replacement. Find $P(\text{exactly one Ace})$.

CHAPTER 04

Descriptive Statistics

4.1 Frequency Distributions

Definition — Frequency Distribution

A **frequency distribution** organises raw data into classes. For each class $[L_i, U_i)$:

- **Midpoint:** $m_i = (L_i + U_i)/2$
- **Frequency** f_i : number of observations in class i
- **Cumulative Frequency** $F_i = \sum_{j \leq i} f_j$
- **Relative Frequency:** f_i/n

4.2 Measures of Central Tendency

Definition — Arithmetic Mean

Ungrouped: $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

Grouped: $\bar{x} = \frac{\sum f_i m_i}{\sum f_i}$ where m_i is the class midpoint.

Properties: (i) $\sum (x_i - \bar{x}) = 0$; (ii) $\sum (x_i - \bar{x})^2$ is minimised by $\bar{x}$; (iii) affected by outliers.

Definition — Median

The **median** is the middle value of ordered data. For grouped data:

$$\text{Median} = L + \frac{h}{f} \left(\frac{N}{2} - F \right)$$

where L = lower boundary of median class, h = class width, f = frequency of median class, F = cumulative frequency before median class, $N = \sum f$.

Median class: the class containing the $N/2$ -th observation.

Definition — Mode

The **mode** is the most frequently occurring value. For grouped data:

$$\text{Mode} = L + h \cdot \frac{f_1 - f_0}{(f_1 - f_0) + (f_1 - f_2)}$$

where f_1 = modal class frequency, f_0 = frequency before, f_2 = frequency after.

4.3 Measures of Dispersion

Definition — Mean Deviation

$$\text{MD from mean} = \frac{\sum f_i |m_i - \bar{x}|}{\sum f_i}$$

Coefficient of Mean Deviation: $\text{Cmd} = \frac{\text{MD}}{\bar{x}}$

Definition — Variance and Standard Deviation

Ungrouped (sample):

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \frac{\sum x_i^2 - n\bar{x}^2}{n - 1}$$

Grouped (sample):

$$s^2 = \frac{\sum f_i (m_i - \bar{x})^2}{\sum f_i - 1}$$

Standard deviation $s = \sqrt{s^2}$. The division by $n - 1$ (not n) makes s^2 an *unbiased* estimator of σ^2 .

Definition — Coefficient of Variation and Coefficient of Standard Deviation

$$\text{CV} = \frac{s}{\bar{x}} \times 100\% \quad \text{Coefficient of SD} = \frac{s}{\bar{x}}$$

These are **dimensionless** measures that allow comparison of variability across datasets with different units or different means.

Lower CV $\Rightarrow$ more homogeneous, more consistent dataset.

Example 4.1 — Full Descriptive Analysis from Grouped Data

Marks of 50 students in a statistics test:

Class	f_i	m_i	F_i	$f_i m_i$	$f_i (m_i - \bar{x})^2$
10–20	3	15	3	45	—
20–30	7	25	10	175	—
30–40	15	35	25	525	—
40–50	12	45	37	540	—
50–60	8	55	45	440	—
60–70	5	65	50	325	—
Total	50			2050	

Mean: $\bar{x} = 2050/50 = 41.0$

Median ($N/2 = 25$): 25th obs falls in 30–40 ($F = 10$, $f = 15$).

$$\text{Median} = 30 + \frac{10}{15}(25 - 10) = 30 + 10 = \mathbf{40.0}$$

Q_1 ($N/4 = 12.5$): class 30–40 ($F = 10$, $f = 15$).

$$Q_1 = 30 + \frac{10}{15}(12.5 - 10) = 30 + 1.67 = \mathbf{31.67}$$

Q_3 ($3N/4 = 37.5$): class 40–50 ($F = 25$, $f = 12$).

$$Q_3 = 40 + \frac{10}{12}(37.5 - 25) = 40 + 10.42 = \mathbf{50.42}$$

Now computing $f_i(m_i - 41)^2$: $3(676) + 7(256) + 15(36) + 12(16) + 8(196) + 5(576) = 2028 + 1792 + 540 + 192 + 1568 + 2880 = \mathbf{8200}$

$s^2 = 8200/49 = 167.35$; $s = \sqrt{167.35} = \mathbf{12.94}$; $CV = 12.94/41 \times 100 = \mathbf{31.6\%}$

4.4 Measures of Position

Definition — General Interpolation Formula

For quartiles ($k = 4$), deciles ($k = 10$), percentiles ($k = 100$):

$$M_r = L + \frac{h}{f} \left(\frac{rN}{k} - F \right)$$

- r = required number (e.g. $r = 1$ for Q_1 , $r = 3$ for D_3 , $r = 75$ for P_{75})
- L = lower class boundary of required class
- h = class width; f = class frequency; F = cumulative frequency before required class

Example 4.2 — Deciles and Percentiles

Using the data from Example 4.1 ($N = 50$):

D_3 ($3N/10 = 15$): class 30–40 ($F = 10$, $f = 15$).

$$D_3 = 30 + \frac{10}{15}(15 - 10) = 30 + 3.33 = \mathbf{33.33}$$

D_7 ($7N/10 = 35$): class 40–50 ($F = 25$, $f = 12$).

$$D_7 = 40 + \frac{10}{12}(35 - 25) = 40 + 8.33 = \mathbf{48.33}$$

P_{60} ($60N/100 = 30$): class 40–50 ($F = 25$, $f = 12$).

$$P_{60} = 40 + \frac{10}{12}(30 - 25) = 40 + 4.17 = \mathbf{44.17}$$

P_{85} ($85N/100 = 42.5$): class 50–60 ($F = 37$, $f = 8$).

$$P_{85} = 50 + \frac{10}{8}(42.5 - 37) = 50 + 6.875 = \mathbf{56.88}$$

Example 4.3 — Comparing Consistency of Two Groups

Test scores of two sections (30 students each):

Section	$\bar{x}$	s	CV	MD	Cmd
A (Morning)	65.4	8.2	12.5%	6.8	0.104
B (Evening)	72.1	14.6	20.2%	11.9	0.165

Conclusion: Section B has a higher mean but Section A is more consistent — lower CV and Cmd indicate less variability relative to the mean. Section A students are more homogeneous in performance.

Practice Problems

- Q1.** Given $\bar{x} = 45$, $s = 9$, find the CV and interpret it.
- Q2.** For grouped data with $N = 80$: find the median class if cumulative frequencies are 12, 28, 55, 70, 80.
- Q3.** Two machines: Machine A ($\bar{x} = 200$ units, $s = 15$); Machine B ($\bar{x} = 185$ units, $s = 10$). Which is more consistent?
- Q4.** Compute Q_2 , D_5 , and P_{50} for any dataset and comment on the relationship.

CHAPTER 05

Sampling Distributions

Historical Background

The concept of a sampling distribution was implicit in the work of **Abraham de Moivre** (1733), who proved the normal approximation to the binomial. **Carl Friedrich Gauss** (1809) derived the normal distribution in the context of measurement errors. The χ^2 distribution was characterised by **Karl Pearson** in 1900. **William S. Gosset** (writing as “Student”, 1908) derived the t -distribution while working at Guinness Brewery — he needed to handle small samples in quality control. **R. A. Fisher** introduced the F -distribution (1924) and proved the independence of $\bar{X}$ and s^2 for normal populations. The Central Limit Theorem in its modern form is due to **Aleksandr Lyapunov** (1901) and **Jarl Waldemar Lindeberg** (1922).

Definition — Sampling Distribution

The **sampling distribution** of a statistic $T = T(X_1, \dots, X_n)$ is its probability distribution over all possible samples of size n from the population.

For the sample mean $\bar{X}$:

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

The standard error **decreases** as n increases — larger samples give more precise estimates.

5.1 Normal Distribution

Definition — Normal Distribution $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

Standard Normal: $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$. CDF: $\Phi(z) = P(Z \leq z)$.

Properties: symmetric about μ ; mean = median = mode = μ ; total area = 1; bell-shaped.

Empirical Rule (68-95-99.7):

$P(\mu - \sigma < X < \mu + \sigma) \approx 68\%$; $P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 95\%$; $P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 99.7\%$

Example 5.1 — Normal Probability Calculations

$X \sim N(60, 100)$ (mean 60, $\sigma = 10$). Find:

(a) $P(X < 75) = P\left(Z < \frac{75-60}{10}\right) = P(Z < 1.5) = \Phi(1.5) = 0.9332$

(b) $P(50 < X < 70) = P(-1 < Z < 1) = \Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = \mathbf{0.6827}$

(c) $P(X > 80) = P(Z > 2) = 1 - 0.9772 = \mathbf{0.0228}$

(d) Find x such that $P(X < x) = 0.90$: $Z_{0.90} = 1.282$, so $x = 60 + 1.282(10) = \mathbf{72.82}$

5.2 Chi-Square Distribution**Definition — χ^2 Distribution**

If $Z_1, \dots, Z_k \stackrel{iid}{\sim} N(0, 1)$ then:

$$\chi^2 = \sum_{i=1}^k Z_i^2 \sim \chi^2(k)$$

PDF: $f(x) = \frac{x^{k/2-1} e^{-x/2}}{2^{k/2} \Gamma(k/2)}, x > 0.$

Mean = k , Variance = $2k$. **Right-skewed**; approaches normality as $k \rightarrow \infty$.

Key result: If $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, then:

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

and $\bar{X}$ and s^2 are **independent** (Fisher's theorem, 1925).

Derivation — Why $(n-1)$ Degrees of Freedom?

$\sum (X_i - \bar{X})^2$ involves n terms subject to the constraint $\sum (X_i - \bar{X}) = 0$, leaving $n-1$ free terms. Each standardised free term contributes one chi-square degree of freedom. The constraint “costs” one degree of freedom.

5.3 Student's t -Distribution

Definition — t -Distribution $t(\nu)$

If $Z \sim N(0, 1)$ and $V \sim \chi^2(\nu)$ are **independent**:

$$T = \frac{Z}{\sqrt{V/\nu}} \sim t(\nu)$$

$$\text{PDF: } f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}$$

Properties: symmetric about 0; heavier tails than $N(0, 1)$; as $\nu \rightarrow \infty$, $t(\nu) \rightarrow N(0, 1)$.

Practical form: When σ is unknown and $X_i \sim N(\mu, \sigma^2)$:

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1)$$

Derivation — Derivation of the t -Statistic

Since $\bar{X} \sim N(\mu, \sigma^2/n)$: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$.

And $V = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$, independent of Z .

Therefore: $T = \frac{Z}{\sqrt{V/(n-1)}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{(n-1)s^2/\sigma^2/(n-1)}} = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1)$. □

5.4 F -Distribution

Definition — F -Distribution $F(m, n)$

If $U \sim \chi^2(m)$ and $V \sim \chi^2(n)$ are independent:

$$F = \frac{U/m}{V/n} \sim F(m, n)$$

Mean = $n/(n-2)$ for $n > 2$. Right-skewed. **Key property:** $1/F \sim F(n, m)$.

Use: $F = s_1^2/s_2^2 \sim F(n_1 - 1, n_2 - 1)$ to compare two population variances.

5.5 Central Limit Theorem

Theorem — Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be iid with mean μ and finite variance σ^2 . Then:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty$$

equivalently $\bar{X} \stackrel{\text{approx}}{\sim} N(\mu, \sigma^2/n)$ for large n .

Proof sketch (Lindeberg-Lévy): The moment generating function (MGF) of $\bar{X}$ standardised converges pointwise to $e^{t^2/2}$, the MGF of $N(0, 1)$. Convergence of MGFs implies convergence in distribution. $\square$

Rule of thumb: $n \geq 30$ usually sufficient. For proportions, need $np \geq 5$ and $n(1-p) \geq 5$.

Example 5.2 — CLT in Practice

Incomes in a city: $\mu = \$45000$, $\sigma = \$12000$. A sample of $n = 100$ is taken.

By CLT: $\bar{X} \sim N(45000, 12000^2/100) = N(45000, 1200^2)$.

(a) $P(\bar{X} > 47000) = P\left(Z > \frac{47000 - 45000}{1200}\right) = P(Z > 1.67) = 1 - 0.9525 = \mathbf{0.0475}$

(b) $P(44000 < \bar{X} < 46000) = P\left(\frac{-1000}{1200} < Z < \frac{1000}{1200}\right) = P(-0.83 < Z < 0.83)$
 $= 2\Phi(0.83) - 1 = 2(0.7967) - 1 = \mathbf{0.5934}$

Distribution	Symbol	Parameters	Mean	Variance
Normal	$N(\mu, \sigma^2)$	μ, σ^2	μ	σ^2
Chi-Square	$\chi^2(k)$	k (df)	k	$2k$
Student's t	$t(\nu)$	ν (df)	0 ($\nu > 1$)	$\nu/(\nu - 2)$ ($\nu > 2$)
F	$F(m, n)$	m, n (df)	$n/(n - 2)$	—

CHAPTER 06

Point Estimation

Historical Background

The theory of estimation was systematised by **R. A. Fisher**, who introduced *maximum likelihood estimation* (1912, 1922), *sufficiency* (1920), and the *information inequality* (1925). **C. R. Rao** and **Harald Cramér** independently derived the lower bound for the variance of unbiased estimators (1945–1946). **David Blackwell** proved his conditioning theorem (1947), and **Erich Lehmann** with **Henry Scheffé** combined it with completeness to characterise the unique minimum-variance unbiased estimator (1950).

Definition — Point Estimator and Estimate

A **point estimator** $\hat{\theta} = T(X_1, \dots, X_n)$ is a statistic used to estimate the unknown parameter θ . A specific computed value $\hat{\theta} = T(x_1, \dots, x_n)$ is a **point estimate**.

6.1 Properties of Estimators

Definition — Unbiasedness

$\hat{\theta}$ is **unbiased** for θ if $E(\hat{\theta}) = \theta$ for all θ .

$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$. An unbiased estimator has zero bias.

Examples:

$\bar{X}$ is unbiased for μ : $E(\bar{X}) = \mu$. ✓

$s^2 = \sum (X_i - \bar{X})^2 / (n - 1)$ is unbiased for σ^2 : $E(s^2) = \sigma^2$. ✓

$\hat{\sigma}_{MLE}^2 = \sum (X_i - \bar{X})^2 / n$ is *biased*: $E(\hat{\sigma}_{MLE}^2) = \frac{n-1}{n} \sigma^2 \neq \sigma^2$. ✗

Definition — Mean Squared Error (MSE)

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

For unbiased estimators: $\text{MSE} = \text{Variance}$. The MSE captures both bias and variance simultaneously.

A **biased** estimator can have smaller MSE than an unbiased one if its variance is sufficiently small.

Definition — Consistency

$\hat{\theta}_n$ is **consistent** for θ if $\hat{\theta}_n \xrightarrow{P} \theta$, i.e. $\forall \varepsilon > 0: \lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \varepsilon) = 0$.

Sufficient condition: $\text{MSE}(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$.

$\bar{X}$ is consistent: $\text{MSE}(\bar{X}) = \sigma^2/n \rightarrow 0$.

Definition — Relative Efficiency

Among unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$, the **relative efficiency** of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is:

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

$\hat{\theta}_1$ is **more efficient** if $\text{eff} > 1$. The most efficient estimator is called the **UMVUE** (Uniformly Minimum Variance Unbiased Estimator).

Definition — Sufficiency and Factorization Theorem

A statistic $T(\mathbf{X})$ is **sufficient** for θ if the conditional distribution of $\mathbf{X}$ given $T = t$ does not depend on θ — T contains *all the information* about θ in the sample.

Neyman Factorization Theorem: T is sufficient for θ iff the joint pdf/pmf factors as:

$$f(\mathbf{x}; \theta) = g(T(\mathbf{x}), \theta) \cdot h(\mathbf{x})$$

where g depends on the data *only through* T , and h does not depend on θ .

Example 6.1 — Sufficiency via Factorization

Bernoulli: $X_i \sim \text{Bernoulli}(p)$. Joint pmf:

$$f(\mathbf{x}; p) = p^{\sum x_i} (1-p)^{n-\sum x_i} = \underbrace{p^T (1-p)^{n-T}}_{g(T,p)} \cdot \underbrace{1}_{h(\mathbf{x})}$$

where $T = \sum X_i$. By factorization, $T = \sum X_i$ is **sufficient** for p .

Normal: For $X_i \sim N(\mu, \sigma^2)$ with σ^2 known, $\bar{X}$ is sufficient for μ .

For both μ and σ^2 unknown, $(\bar{X}, s^2)$ jointly form a **complete sufficient** statistic.

6.2 Cramer-Rao Lower Bound

Theorem — Cramer-Rao Lower Bound (CRLB)

Under regularity conditions, for any unbiased estimator $\hat{\theta}$ of θ :

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

where the **Fisher Information** is:

$$I(\theta) = E \left[\left(\frac{\partial \ln f(X; \theta)}{\partial \theta} \right)^2 \right] = -E \left[\frac{\partial^2 \ln f(X; \theta)}{\partial \theta^2} \right]$$

An unbiased estimator achieving the CRLB is called **efficient** (and is automatically the UMVUE).

Derivation — Proof of CRLB

Under regularity (interchange differentiation and integration): $E \left[\frac{\partial \ln f}{\partial \theta} \right] = 0$ (score has mean zero).

By Cauchy-Schwarz: $\left[\text{Cov} \left(\hat{\theta}, \frac{\partial \ln f}{\partial \theta} \right) \right]^2 \leq \text{Var}(\hat{\theta}) \cdot \text{Var} \left(\frac{\partial \ln f}{\partial \theta} \right)$

The left side equals $[\partial E(\hat{\theta}) / \partial \theta]^2 = [1]^2 = 1$ (since $\hat{\theta}$ is unbiased).

The right side gives $\text{Var}(\hat{\theta}) \cdot nI(\theta) \geq 1$, so $\text{Var}(\hat{\theta}) \geq 1/(nI(\theta))$. □

Example 6.2 — CRLB for Normal Mean

$X \sim N(\mu, \sigma^2)$, σ^2 known. $\ln f = -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}$.

$\frac{\partial^2 \ln f}{\partial \mu^2} = -\frac{1}{\sigma^2}$, so $I(\mu) = 1/\sigma^2$.

CRLB = σ^2/n . Since $\text{Var}(\bar{X}) = \sigma^2/n$, $\bar{X}$ is **efficient** for μ .

Example 6.3 — CRLB for Poisson

$X \sim \text{Poisson}(\lambda)$. $\ln f = x \ln \lambda - \lambda - \ln(x!)$.

$\frac{\partial^2 \ln f}{\partial \lambda^2} = -x/\lambda^2$, so $I(\lambda) = E(X)/\lambda^2 = 1/\lambda$.

CRLB = λ/n . Since $\text{Var}(\bar{X}) = \lambda/n$, $\bar{X}$ is **efficient** for λ .

6.3 Rao-Blackwell and Lehmann-Scheffé Theorems

Theorem — Rao-Blackwell Theorem (1947)

Let $\hat{\theta}$ be any unbiased estimator of θ and T a sufficient statistic. Define:

$$\hat{\theta}^* = E(\hat{\theta} | T)$$

Then $\hat{\theta}^*$ is: (i) a statistic (function of data only); (ii) unbiased: $E(\hat{\theta}^*) = \theta$; (iii) at least as efficient: $\text{Var}(\hat{\theta}^*) \leq \text{Var}(\hat{\theta})$.

Proof of (iii): By Eve's law: $\text{Var}(\hat{\theta}) = E[\text{Var}(\hat{\theta} | T)] + \text{Var}[E(\hat{\theta} | T)] = E[\text{Var}(\hat{\theta} | T)] + \text{Var}(\hat{\theta}^*) \geq \text{Var}(\hat{\theta}^*)$. $\square$

Theorem — Lehmann-Scheffé Theorem (1950)

If T is a **complete sufficient** statistic and $\hat{\theta}^* = E(\hat{\theta} | T)$ is unbiased for θ , then $\hat{\theta}^*$ is the **unique UMVUE** of θ .

6.4 Method of Moments (MOM)

Definition — Method of Moments

Population r -th moment: $\mu'_r = E(X^r)$

Sample r -th moment: $m'_r = \frac{1}{n} \sum_{i=1}^n X_i^r$

MOM: Set $\mu'_r = m'_r$ for $r = 1, 2, \dots, k$ (where k = number of parameters) and solve for the parameters.

Example 6.4 — MOM Estimators

(a) **Normal** $N(\mu, \sigma^2)$: Two parameters, use two moments.

$$\mu'_1 = \mu \Rightarrow \hat{\mu} = \bar{X}. \quad \mu'_2 = \sigma^2 + \mu^2 \Rightarrow \hat{\sigma}^2 = m'_2 - \bar{X}^2 = \frac{1}{n} \sum X_i^2 - \bar{X}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$$

(b) **Exponential** $\text{Exp}(\lambda)$: $E(X) = 1/\lambda \Rightarrow \hat{\lambda}_{MOM} = 1/\bar{X}$.

(c) **Gamma** (α, β) : $E(X) = \alpha\beta$ and $\text{Var}(X) = \alpha\beta^2$.

$$\hat{\beta} = s^2/\bar{X}, \quad \hat{\alpha} = \bar{X}^2/s^2.$$

Advantages

- Simple to compute — always produces an estimator
- No distributional assumptions beyond moment existence
- Consistent estimators under mild conditions

Limitations

- Not always efficient (can have larger variance than MLE)
- Estimates may fall outside the parameter space (e.g. negative variance estimate)
- Ignores the full likelihood information

6.5 Maximum Likelihood Estimation (MLE)

Definition — Likelihood and MLE

Given observations $x_1, \dots, x_n$, the **likelihood function** is:

$$L(\theta) = L(\theta; \mathbf{x}) = \prod_{i=1}^n f(x_i; \theta)$$

The **log-likelihood** is $\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(x_i; \theta)$.

The **MLE** $\hat{\theta}$ maximises $L(\theta)$ (equivalently $\ell(\theta)$). **Score equation:** $\frac{\partial \ell}{\partial \theta} = 0$.

Properties of MLE:

- **Consistent:** $\hat{\theta}_{MLE} \xrightarrow{P} \theta$ as $n \rightarrow \infty$
- **Asymptotically Normal:** $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$
- **Asymptotically Efficient:** achieves CRLB as $n \rightarrow \infty$
- **Invariance:** if $\hat{\theta}$ is MLE of θ , then $g(\hat{\theta})$ is MLE of $g(\theta)$

Example 6.5 — MLE for Binomial, Exponential, Normal

(a) **Binomial:** $X \sim B(n, p)$. $\ell(p) = x \ln p + (n - x) \ln(1 - p)$.

$$d\ell/dp = x/p - (n - x)/(1 - p) = 0 \Rightarrow \boxed{\hat{p} = x/n}$$

(b) **Exponential:** $f(x; \lambda) = \lambda e^{-\lambda x}$. $\ell(\lambda) = n \ln \lambda - \lambda \sum x_i$.

$$d\ell/d\lambda = n/\lambda - \sum x_i = 0 \Rightarrow \boxed{\hat{\lambda} = 1/\bar{x}}$$

(c) **Normal** (μ, σ^2 both unknown): $\ell = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$

$$\partial\ell/\partial\mu = 0 \Rightarrow \hat{\mu} = \bar{x}; \quad \partial\ell/\partial\sigma^2 = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 \text{ (biased)}$$

(d) **Uniform** $U(0, \theta)$: $L(\theta) = \theta^{-n}$ for $\theta \geq x_{(n)}$. L is decreasing in θ , so minimize over valid

$$\text{values: } \boxed{\hat{\theta} = X_{(n)} = \max(X_i)}.$$

Note: $E(X_{(n)}) = \frac{n}{n+1}\theta$ — biased. Unbiased version: $\frac{n+1}{n}X_{(n)}$.

Practice Problems

- Q1.** Find MLE of p for the Geometric distribution $P(X = x) = p(1 - p)^{x-1}$.
- Q2.** Show $\bar{X}$ is unbiased and efficient for μ when $X_i \sim N(\mu, \sigma^2)$ with σ^2 known.
- Q3.** Compute CRLB for estimating λ in the Exponential distribution.
- Q4.** Find MOM and MLE estimators for θ in $f(x; \theta) = (\theta + 1)x^\theta$, $0 < x < 1$.

CHAPTER 07

Interval Estimation

Definition — Confidence Interval

A $(1 - \alpha) \times 100\%$ **confidence interval (CI)** for θ is a random interval (L, U) where L and U are statistics satisfying:

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Interpretation: If we constructed such intervals from infinitely many samples, $(1 - \alpha) \times 100\%$ of them would contain the true parameter. It is *incorrect* to say “there is $(1 - \alpha)$ probability that θ lies in (L, U) ” for a specific realised interval — θ is fixed (frequentist view).

$(1 - \alpha)$ = **confidence level**; α = **significance level**; $E = z_{\alpha/2} \cdot \text{SE}$ = **margin of error**.

Definition — Pivotal Quantity Method

A **pivot** $Q(\mathbf{X}, \theta)$ is a function of the data and θ whose distribution does not depend on θ .

Method: Find Q ; find c_1, c_2 with $P(c_1 \leq Q \leq c_2) = 1 - \alpha$; rearrange to isolate θ .

7.1 CI for Mean

Derivation — Derivation of CI for μ (σ known)

Pivot: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$.

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$\text{Rearranging: } P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Formula — CI for μ

σ **known** (Z-interval): $\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

σ **unknown** (T-interval): $\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$

Common z values: $z_{0.05} = 1.645$ (90%); $z_{0.025} = 1.960$ (95%); $z_{0.005} = 2.576$ (99%).

Example 7.1 — CI for Mean

Known σ : A machine fills bags; $\sigma = 5\text{g}$. Sample: $n = 36$, $\bar{x} = 502\text{g}$. Find 99% CI.
 $502 \pm 2.576 \times 5/6 = 502 \pm 2.147 = (499.85, 504.15)\text{g}$

Unknown σ : IQ scores: $n = 25$, $\bar{x} = 108.5$, $s = 12.3$. Find 95% CI.

$$t_{0.025,24} = 2.064. \quad 108.5 \pm 2.064 \times 12.3/5 = 108.5 \pm 5.077 = (\mathbf{103.42}, \mathbf{113.58})$$

7.2 Sample Size Determination

Formula — Required Sample Size

To estimate μ within margin of error E at confidence $(1 - \alpha)$:

$$n = \left\lceil \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 \right\rceil$$

To estimate proportion p within E :

$$n = \left\lceil \frac{z_{\alpha/2}^2 p(1-p)}{E^2} \right\rceil \quad (\text{use } p = 0.5 \text{ if unknown, gives largest } n)$$

Always round **up** to the nearest integer.

Example 7.2 — Sample Size

(a) Estimate mean exam score within $E = 3$ marks at 95%, $\sigma = 15$:

$$n = (1.96 \times 15/3)^2 = (9.8)^2 = 96.04 \Rightarrow n = \mathbf{97}$$

(b) Estimate proportion who pass within $E = 0.04$ at 99%, p unknown:

$$n = \lceil 2.576^2 \times 0.25/0.0016 \rceil = \lceil 6.6306 \times 156.25 \rceil = \lceil 1036 \rceil = \mathbf{1036}$$

7.3 CI for Variance and Proportion

Formula — CI for σ^2 and p

Variance: Pivot $V = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$.

$$\left(\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2} \right)$$

Proportion (large n): $\hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$ (valid when $n\hat{p} \geq 5$, $n(1-\hat{p}) \geq 5$)

7.4 CI for Difference of Two Means

Formula — Two-Sample CIs

Equal variances ($\sigma_1^2 = \sigma_2^2$ assumed):

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \quad s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$$

Unequal variances (Welch):

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, \nu} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \quad \nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

Example 7.3 — Two-Sample CI

Diet A ($n_1 = 12$, $\bar{x}_1 = 5.2$ kg lost, $s_1 = 1.8$) vs Diet B ($n_2 = 10$, $\bar{x}_2 = 3.7$ kg, $s_2 = 2.1$). Assuming equal variances, find 95% CI for $\mu_1 - \mu_2$.

$$s_p^2 = \frac{11(3.24) + 9(4.41)}{20} = \frac{35.64 + 39.69}{20} = \frac{75.33}{20} = 3.767; \quad s_p = 1.941$$

$$t_{0.025, 20} = 2.086. \quad (5.2 - 3.7) \pm 2.086 \times 1.941 \sqrt{1/12 + 1/10} = 1.5 \pm 2.086 \times 0.856 = 1.5 \pm 1.786$$

CI: $(-0.286, 3.286)$ — since 0 is included, the difference is not significant at 5%.

CHAPTER 08

Hypothesis Testing

Historical Background

The modern framework of hypothesis testing was developed by **Jerzy Neyman** and **Egon Pearson** (son of Karl Pearson) in a landmark series of papers (1928–1933). They introduced the concepts of Type I and Type II errors, power, and the likelihood ratio principle. Their work stood in philosophical tension with **R. A. Fisher**'s significance testing framework — Fisher rejected the concept of Type II error and power. Today, both approaches are taught together. The p -value concept is due to Fisher, while the formal error-control framework is due to Neyman-Pearson. **Karl Pearson**'s chi-square test (1900) was one of the first formal statistical tests.

8.1 Framework of Hypothesis Testing

Definition — Null and Alternative Hypotheses

H_0 (**null hypothesis**): the default claim, assumed true until evidence rejects it.

H_1 (**alternative hypothesis**): the research claim we seek to support.

Two-tailed: $H_1 : \theta \neq \theta_0$

Right-tailed: $H_1 : \theta > \theta_0$

Left-tailed: $H_1 : \theta < \theta_0$

Definition — Type I Error, Type II Error, Power

Decision	H_0 True	H_0 False
Reject H_0	Type I Error (α)	Correct (Power = $1 - \beta$)
Fail to Reject H_0	Correct	Type II Error (β)

$\alpha = P(\text{Type I}) = P(\text{reject } H_0 \mid H_0 \text{ true}) = \text{significance level (set by researcher)}$.

$\beta = P(\text{Type II}) = P(\text{fail to reject } H_0 \mid H_0 \text{ false})$.

Power = $1 - \beta = P(\text{reject } H_0 \mid H_0 \text{ false})$. We want high power and small α .

Trade-off: For fixed n , decreasing α increases β (and decreases power). Increasing n reduces both.

Definition — p -value

The **p -value** is the probability, under H_0 , of obtaining a test statistic at least as extreme as the observed value.

Decision rule: Reject H_0 iff $p\text{-value} < \alpha$.

Common misinterpretation: The p -value is *not* the probability that H_0 is true. It is a measure of the evidence *against* H_0 in the data.

8.2 One-Sample Tests for Means**Formula — z -Test and t -Test**

One-sample z -test (σ known): $H_0 : \mu = \mu_0$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1). \text{ Reject if } |Z| > z_{\alpha/2} \text{ (two-tailed).}$$

One-sample t -test (σ unknown): $H_0 : \mu = \mu_0$

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t(n-1). \text{ Reject if } |T| > t_{\alpha/2, n-1} \text{ (two-tailed).}$$

Example 8.1 — One-Sample t -Test (Full Solution)

A manufacturer claims the average life of a bulb is 1000 hours. A sample of $n = 20$ bulbs gives $\bar{x} = 980$ hours and $s = 60$ hours. Test at $\alpha = 0.05$.

$H_0 : \mu = 1000$ vs $H_1 : \mu \neq 1000$ (two-tailed).

$$T = \frac{980 - 1000}{60/\sqrt{20}} = \frac{-20}{13.416} = -1.491$$

$t_{0.025, 19} = 2.093$. Since $|-1.491| < 2.093$: **Fail to reject H_0 .**

$p\text{-value} = 2P(T_{19} < -1.491) \approx 2(0.076) = 0.152 > \alpha = 0.05$. Same conclusion.

Interpretation: The sample provides insufficient evidence to dispute the manufacturer's claim.

8.3 Two-Sample Tests**Formula — Two-Sample and Paired Tests**

Two-sample t (equal variances): $H_0 : \mu_1 = \mu_2$

$$T = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{1/n_1 + 1/n_2}} \sim t(n_1 + n_2 - 2)$$

Paired t -test: $D_i = X_{1i} - X_{2i}$; $H_0 : \mu_D = 0$; $T = \frac{\bar{D}}{s_D/\sqrt{n}} \sim t(n-1)$

Example 8.2 — Paired t -Test

Blood pressure measured before and after a new drug for $n = 10$ patients:

Patient	Before	After	D_i
1	145	138	7
2	152	145	7
3	138	132	6
4	160	150	10
5	142	138	4
6	155	147	8
7	148	142	6
8	150	143	7
9	158	152	6
10	145	140	5
			$\Sigma = 66$

$$\bar{D} = 66/10 = 6.6; \quad s_D = \sqrt{\frac{\sum D_i^2 - 10\bar{D}^2}{9}} = \sqrt{\frac{460 - 435.6}{9}} = \sqrt{2.711} = 1.647$$

$$H_1 : \mu_D > 0 \text{ (right-tailed). } T = 6.6 / (1.647 / \sqrt{10}) = 6.6 / 0.521 = 12.67.$$

$t_{0.05,9} = 1.833$. Since $12.67 > 1.833$: **Reject H_0** . Drug significantly reduces blood pressure.

8.4 Tests for Variance**Formula — χ^2 and F Tests**

One-variance: $H_0 : \sigma^2 = \sigma_0^2; \quad \chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi^2(n-1)$

Two-variance: $H_0 : \sigma_1^2 = \sigma_2^2; \quad F = \frac{s_1^2}{s_2^2} \sim F(n_1-1, n_2-1)$

8.5 Chi-Square Tests**Formula — Goodness of Fit and Independence**

Goodness of Fit: $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2(k-1-p)$

(p = estimated parameters; expected $E_i = nP_i$; require $E_i \geq 5$.)

Test of Independence ($r \times c$ table): $\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2((r-1)(c-1))$

$$E_{ij} = \frac{(\text{Row total}_i) \times (\text{Col total}_j)}{n}$$

Example 8.3 — Test of Independence

Is smoking independent of lung disease? Data from 200 patients:

	Lung Disease	No Disease	Total
Smoker	60	40	100
Non-Smoker	30	70	100
Total	90	110	200

$$E_{11} = 90 \times 100/200 = 45; \quad E_{12} = 110 \times 100/200 = 55; \quad E_{21} = 45; \quad E_{22} = 55$$

$$\chi^2 = \frac{(60 - 45)^2}{45} + \frac{(40 - 55)^2}{55} + \frac{(30 - 45)^2}{45} + \frac{(70 - 55)^2}{55} = 5 + 4.09 + 5 + 4.09 = 18.18$$

$\chi_{0.05,1}^2 = 3.841$. Since $18.18 \gg 3.841$: **Reject** H_0 — smoking and lung disease are *not* independent.

8.6 Neyman-Pearson Lemma**Theorem — Neyman-Pearson Lemma (1933)**

For testing $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ (simple vs simple), the **Most Powerful (MP)** test of size α has rejection region:

$$C = \left\{ \mathbf{x} : \frac{L(\theta_1; \mathbf{x})}{L(\theta_0; \mathbf{x})} \geq k \right\}$$

where $k > 0$ is chosen so that $P(\mathbf{X} \in C \mid \theta_0) = \alpha$.

This is the **Uniformly Most Powerful (UMP)** test when the rejection region does not depend on θ_1 for all $\theta_1 \in H_1$ (typically for one-sided tests with monotone likelihood ratio families).

Example 8.4 — Neyman-Pearson Lemma

$X_1, \dots, X_n \sim \text{Bernoulli}(p)$. Test $H_0 : p = 0.3$ vs $H_1 : p = 0.6$ at $\alpha = 0.05$, $n = 10$.

$$\text{LR} = \frac{(0.6)^T (0.4)^{n-T}}{(0.3)^T (0.7)^{n-T}} = \left(\frac{0.6}{0.3} \right)^T \left(\frac{0.4}{0.7} \right)^{n-T}, \text{ which is increasing in } T = \sum X_i.$$

MP test rejects for large T . Reject H_0 if $T \geq c$ where $P(T \geq c \mid p = 0.3) \leq 0.05$.

Under H_0 : $T \sim B(10, 0.3)$. $P(T \geq 6) = 0.047 < 0.05$. **Reject if** $T \geq 6$.

Practice Problems

- Q1.** A sample of 64 observations gives $\bar{x} = 52$, $\sigma = 16$. Test $H_0 : \mu = 50$ at $\alpha = 0.05$ (two-tailed).
- Q2.** A die is rolled 120 times. Expected: 20 each. Observed: 16, 22, 18, 24, 20, 20. Test fairness.
- Q3.** Find the power of the test in Q1 if the true mean is $\mu = 54$.
- Q4.** Write down the test for $H_0 : \sigma_1^2 = \sigma_2^2$ and state the rejection region.

CHAPTER 09

Regression & Correlation

Historical Background

Francis Galton coined the term “regression” in 1886 while studying the relationship between the heights of parents and their children — he observed that children’s heights “regressed to the mean.” His student **Karl Pearson** formalised the correlation coefficient (1896). **R. A. Fisher** developed the analysis of variance framework for regression (1922). The **Gauss-Markov Theorem**, named after Gauss (1821) and Markov (1900), establishes that OLS estimators are the Best Linear Unbiased Estimators (BLUE).

Definition — Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, \dots, n$$

Assumptions (LINE):

1. **Linearity:** $E(Y | X) = \beta_0 + \beta_1 X$ (true relationship is linear)
2. **Independence:** ε_i are independent
3. **Normality:** $\varepsilon_i \sim N(0, \sigma^2)$
4. **Equal variance (homoscedasticity):** $\text{Var}(\varepsilon_i) = \sigma^2$ (constant)

β_0 = population intercept (value of Y when $X = 0$)

β_1 = population slope (change in $E(Y)$ per unit increase in X)

9.1 OLS Estimation

Derivation — Derivation of OLS Estimators

Minimise the Sum of Squared Errors: $SSE = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$.

Normal equations ($\partial SSE / \partial \hat{\beta}_0 = 0$ and $\partial SSE / \partial \hat{\beta}_1 = 0$):

$$\sum Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum X_i \quad \text{and} \quad \sum X_i Y_i = \hat{\beta}_0 \sum X_i + \hat{\beta}_1 \sum X_i^2$$

Solving: $\boxed{\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}}, \quad \boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}} \quad \square$

where $S_{XX} = \sum (X_i - \bar{X})^2 = \sum X_i^2 - n\bar{X}^2$, $S_{XY} = \sum (X_i - \bar{X})(Y_i - \bar{Y}) = \sum X_i Y_i - n\bar{X}\bar{Y}$.

Theorem — Gauss-Markov Theorem

Under the LINE assumptions, the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are **BLUE** (Best Linear Unbiased Estimators): among all linear unbiased estimators of β_0 and β_1 , they have the smallest variance.

Example 9.1 — Complete OLS Regression Analysis

Study hours (X) vs test scores (Y) for 6 students:

i	X	Y	X^2	Y^2	XY
1	1	45	1	2025	45
2	2	55	4	3025	110
3	4	70	16	4900	280
4	5	75	25	5625	375
5	7	85	49	7225	595
6	8	90	64	8100	720
Σ	27	420	159	30900	2125

$\bar{X} = 4.5$, $\bar{Y} = 70$. $S_{XX} = 159 - 6(20.25) = 159 - 121.5 = 37.5$. $S_{XY} = 2125 - 6(4.5)(70) = 2125 - 1890 = 235$.

$S_{YY} = 30900 - 6(70)^2 = 30900 - 29400 = 1500$.

$\hat{\beta}_1 = 235/37.5 = 6.267$; $\hat{\beta}_0 = 70 - 6.267(4.5) = 70 - 28.2 = 41.8$

Regression line: $\hat{Y} = 41.8 + 6.267X$ (each extra hour of study ≈ 6.3 more marks)

Example 9.1 (continued) — Residuals, ANOVA, and R^2

Predicted values and residuals:

i	Y_i	$\hat{Y}_i$	$e_i = Y_i - \hat{Y}_i$
1	45	48.1	-3.1
2	55	54.3	0.7
3	70	66.9	3.1
4	75	73.1	1.9
5	85	85.7	-0.7
6	90	92.0	-2.0

ANOVA for Regression: $SSR = \hat{\beta}_1 S_{XY} = 6.267 \times 235 = 1472.7$; $SSE = 1500 - 1472.7 = 27.3$; $SST = 1500$.

$s^2 = SSE/(n - 2) = 27.3/4 = 6.83$; $R^2 = 1472.7/1500 = \mathbf{0.982}$ — 98.2% of variation explained.

9.2 Inference in Regression

Formula — Standard Errors and t -Tests

$$\text{SE}(\hat{\beta}_1) = \sqrt{s^2/S_{XX}}; \quad \text{SE}(\hat{\beta}_0) = s\sqrt{1/n + \bar{X}^2/S_{XX}}$$

Test $H_0 : \beta_1 = 0$: $T = \hat{\beta}_1/\text{SE}(\hat{\beta}_1) \sim t(n-2)$.

CI for β_1 : $\hat{\beta}_1 \pm t_{\alpha/2, n-2} \cdot \text{SE}(\hat{\beta}_1)$

Formula — Prediction Interval at $X = x^*$

$$\hat{Y}^* \pm t_{\alpha/2, n-2} \cdot s \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{X})^2}{S_{XX}}}$$

The extra “1” under the square root reflects variability in a *single new observation*, not just the mean response.

9.3 Pearson and Spearman Correlation

Definition — Pearson Correlation Coefficient

$$r = \frac{S_{XY}}{\sqrt{S_{XX} \cdot S_{YY}}} \in [-1, 1]$$

$r = +1$: perfect positive linear; $r = -1$: perfect negative linear; $r = 0$: no linear association.

$r^2 = R^2$ in simple regression.

Test $H_0 : \rho = 0$: $T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t(n-2)$.

Definition — Spearman Rank Correlation

Used when data is ordinal or when normality is questionable. Assign ranks R_{X_i}, R_{Y_i} ; compute $d_i = R_{X_i} - R_{Y_i}$:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Example 9.2 — Correlation and Hypothesis Test

From Example 9.1: $r = 235/\sqrt{37.5 \times 1500} = 235/237.2 = 0.991$.

Test $H_0 : \rho = 0$ at $\alpha = 0.05$:

$T = 0.991\sqrt{4}/\sqrt{1-0.982} = 0.991 \times 2/0.134 = 14.79$. $t_{0.025, 4} = 2.776$.

Since $14.79 \gg 2.776$: **Reject H_0** — highly significant positive correlation.

CHAPTER 10

Non-Parametric Tests

Historical Background

Non-parametric statistics emerged in the mid-20th century. **Frank Wilcoxon** introduced his signed-rank test in 1945. **Henry B. Mann** and **D. R. Whitney** developed the rank-sum test in 1947. **William H. Kruskal** and **W. A. Wallis** extended it to multiple groups in 1952. These methods require no distributional assumptions and work with ranks rather than the raw data values, making them robust to outliers and skewness.

Use non-parametric tests when:

- Population distribution is unknown or non-normal
- Data is ordinal (ranked categories)
- Sample size is very small ($n < 10$)
- Outliers are present that cannot be removed
- Parametric assumptions are clearly violated

10.1 Sign Test

Definition — Sign Test

Tests H_0 : population median = M_0 . For each observation record + if $X_i > M_0$, - if $X_i < M_0$; ignore ties.

S = number of + signs. Under H_0 : $S \sim \text{Binomial}(n, 0.5)$.

Large n approximation: $Z = \frac{S - n/2}{\sqrt{n}/2} \sim N(0, 1)$.

Example 10.1 — Sign Test (Full)

Test H_0 : median = 50 for data: 55, 43, 61, 48, 57, 44, 59, 52, 41, 63. $\alpha = 0.05$.

Signs vs 50: +, -, +, -, +, -, +, +, -, +. $S = 6$ positives, $n = 10$.

Under H_0 : $P(S \geq 6)$ from $B(10, 0.5)$.

$P(S \geq 6) = \binom{10}{6}(0.5)^{10} + \dots + \binom{10}{10}(0.5)^{10} = (210 + 120 + 45 + 10 + 1)/1024 = 386/1024 = 0.377$.

Two-tailed $p = 2 \times \min(0.377, 0.623) = 0.754 > \alpha = 0.05$. **Fail to reject H_0 .**

10.2 Wilcoxon Signed-Rank Test

Definition — Wilcoxon Signed-Rank Test

More powerful than sign test (uses both sign and magnitude).

Steps:

1. Compute $D_i = X_i - M_0$; discard zeros; let n' = remaining count
2. Rank $|D_i|$ from 1 (smallest) to n' (largest); average ranks for ties
3. W^+ = sum of ranks where $D_i > 0$; W^- = sum of ranks where $D_i < 0$
4. Test statistic: $W = \min(W^+, W^-)$; compare to critical value from table

Large n' : $Z = \frac{W^+ - n'(n' + 1)/4}{\sqrt{n'(n' + 1)(2n' + 1)/24}} \sim N(0, 1)$.

Note: $W^+ + W^- = n'(n' + 1)/2$ always.

Example 10.2 — Wilcoxon Signed-Rank Test

Test whether median delivery time = 7 days for 8 deliveries: 9, 5, 8, 6, 10, 4, 7, 11.
 $\alpha = 0.05$.

X_i	$D_i = X_i - 7$	$ D_i $	Rank	Signed Rank
9	+2	2	4	+4
5	-2	2	4	-4
8	+1	1	2	+2
6	-1	1	2	-2
10	+3	3	6	+6
4	-3	3	6	-6
7	0	—	—	—
11	+4	4	7	+7

$n' = 7$ (excluding tie). $W^+ = 4 + 2 + 6 + 7 = 19$; $W^- = 4 + 2 + 6 = 12$.
 $W = \min(19, 12) = 12$. Critical value (two-tailed, $n' = 7$, $\alpha = 0.05$): $W_c = 2$.
 Since $12 > 2$: **Fail to reject H_0** . No evidence that median differs from 7.

10.3 Mann-Whitney U Test

Definition — Mann-Whitney U Test

Two independent samples, H_0 : identical distributions.

Steps: Combine and rank all $N = n_1 + n_2$ observations. Let R_1 = rank sum for sample 1.

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1, \quad U_2 = n_1 n_2 - U_1, \quad U = \min(U_1, U_2)$$

Large samples: $Z = \frac{U - n_1 n_2 / 2}{\sqrt{n_1 n_2 (N + 1) / 12}} \sim N(0, 1)$.

10.4 Kruskal-Wallis Test

Definition — Kruskal-Wallis Test

Non-parametric alternative to one-way ANOVA for k independent groups ($k \geq 2$).

Combine all $N = \sum n_i$ observations and rank them. Let R_i = rank sum for group i .

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1) \sim \chi^2(k-1) \text{ under } H_0$$

Reject H_0 if $H > \chi_{\alpha, k-1}^2$.

Test	Parametric Equiv.	H_0	Stat
Sign Test	One-sample t	Median = M_0	$S \sim B(n, 0.5)$
Wilcoxon S-R	One-sample t	Median = M_0	$W = \min(W^+, W^-)$
Mann-Whitney	Two-sample t	Equal distributions	$U = \min(U_1, U_2)$
Kruskal-Wallis	One-way ANOVA	All groups equal	$H \sim \chi^2(k-1)$

CHAPTER 11

Bayesian Inference

Historical Background

Thomas Bayes (1701–1761), an English minister, wrote an essay that was published posthumously in 1763 by his friend Richard Price. **Pierre-Simon Laplace** independently developed and extended Bayesian ideas in the late 18th century. The Bayesian approach fell out of favour for much of the 20th century due to controversy over subjective priors. Its revival began in the 1980s–1990s with the development of computational methods (**MCMC**: Markov Chain Monte Carlo) that made Bayesian calculations tractable for complex models. Today Bayesian methods are fundamental to machine learning, medical decision-making, and scientific inference.

Frequentist vs Bayesian Philosophy:

Frequentist: θ is a fixed unknown constant. Probability = long-run frequency. CIs and p -values are tools for controlling error rates.

Bayesian: θ is a random variable with a probability distribution reflecting our uncertainty. Probability = degree of belief. Prior belief is updated by data to give posterior belief.

11.1 Bayes' Theorem for Parameters

Theorem — Posterior Distribution

$$\underbrace{\pi(\theta | \mathbf{x})}_{\text{posterior}} = \frac{\overbrace{f(\mathbf{x} | \theta)}^{\text{likelihood}} \cdot \overbrace{\pi(\theta)}^{\text{prior}}}{\underbrace{\int f(\mathbf{x} | \theta) \pi(\theta) d\theta}_{\text{marginal likelihood}}} \propto f(\mathbf{x} | \theta) \cdot \pi(\theta)$$

Posterior $\propto$ **Likelihood** $\times$ **Prior**

The marginal likelihood $m(\mathbf{x})$ is a normalising constant that does not depend on θ .

11.2 Conjugate Priors

Definition — Conjugate Prior

A prior $\pi(\theta)$ is **conjugate** to the likelihood $f(\mathbf{x} \mid \theta)$ if the posterior $\pi(\theta \mid \mathbf{x})$ belongs to the same parametric family as the prior. This makes posterior computation analytically tractable.

Likelihood	Param	Conjugate Prior	Posterior
Binomial(n, p)	p	Beta(α, β)	Beta($\alpha + x, \beta + n - x$)
Poisson(λ)	λ	Gamma(α, β)	Gamma($\alpha + \sum x_i, \beta + n$)
Normal (σ^2 known)	μ	$N(\mu_0, \tau_0^2)$	$N(\mu_n, \tau_n^2)$ (see below)
Exponential(λ)	λ	Gamma(α, β)	Gamma($\alpha + n, \beta + \sum x_i$)

Derivation — Normal-Normal Conjugate

$X_i \mid \mu \sim N(\mu, \sigma^2)$, σ^2 known. Prior: $\mu \sim N(\mu_0, \tau_0^2)$.

Posterior precision: $\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$

Posterior mean: $\mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$

μ_n is a **precision-weighted average** of the prior mean and the sample mean. As $n \rightarrow \infty$, $\mu_n \rightarrow \bar{x}$ (data dominates).

11.3 Loss Functions and Bayes Estimators

Definition — Bayes Estimator

The Bayes estimator minimises the expected posterior loss:

Loss Function $L(\hat{\theta}, \theta)$	Bayes Estimator	Name
$(\hat{\theta} - \theta)^2$	$E(\theta \mid \mathbf{x})$	Posterior mean
$ \hat{\theta} - \theta $	Median of $\pi(\theta \mid \mathbf{x})$	Posterior median
$\mathbf{1}(\hat{\theta} \neq \theta)$	Mode of $\pi(\theta \mid \mathbf{x})$	MAP estimator

11.4 Bayesian Credible Interval

Definition — Credible Interval

A $(1 - \alpha) \times 100\%$ **credible interval** (a, b) satisfies:

$$P(a \leq \theta \leq b \mid \mathbf{x}) = 1 - \alpha$$

Direct interpretation: “Given the data and prior, there is a $(1 - \alpha)$ probability that $\theta \in (a, b)$.”

This is a direct probability statement about θ , unlike the frequentist CI.

Example 11.1 — Beta-Binomial Model

A new drug is tested on $n = 20$ patients; $x = 14$ respond positively. Prior belief: $p \sim \text{Beta}(3, 3)$ (reflects moderate prior evidence for effectiveness, centered at 0.5).

Posterior: $p \mid x \sim \text{Beta}(3 + 14, 3 + 6) = \text{Beta}(17, 9)$

Posterior mean (Bayes estimate): $\hat{p} = 17/(17 + 9) = 17/26 \approx \mathbf{0.654}$

MLE: $\hat{p}_{MLE} = 14/20 = 0.70$. The prior pulled the estimate towards 0.5.

95% Credible Interval: Approximately (0.44, 0.83) from the $\text{Beta}(17, 9)$ quantiles.

Interpretation: “Given the data and our prior, we believe there is a 95% probability that the true response rate p lies between 0.44 and 0.83.”

Example 11.2 — Normal-Normal Model

Prior belief: average exam score $\mu \sim N(60, 25)$ (i.e. $\mu_0 = 60$, $\tau_0^2 = 25$). Data: $n = 16$ students, $\bar{x} = 68$, known $\sigma^2 = 64$.

$$\frac{1}{\tau_n^2} = \frac{1}{25} + \frac{16}{64} = 0.04 + 0.25 = 0.29 \Rightarrow \tau_n^2 = 3.448$$

$$\mu_n = 3.448 \left(\frac{60}{25} + \frac{16 \times 68}{64} \right) = 3.448(2.4 + 17) = 3.448 \times 19.4 = \mathbf{66.9}$$

Posterior: $\mu \mid \mathbf{x} \sim N(66.9, 3.448)$. 95% credible interval: $66.9 \pm 1.96\sqrt{3.448} = 66.9 \pm 3.64 = (63.3, 70.5)$.

The posterior mean 66.9 lies between prior mean (60) and sample mean (68), weighted by their precisions.

Practice Problems

- Q1.** Prior: $\lambda \sim \text{Gamma}(2, 1)$. Observe $n = 5$ Poisson counts: 3, 5, 2, 4, 6. Find the posterior and Bayes estimate.
- Q2.** Show that for a Binomial likelihood with Beta prior, as $n \rightarrow \infty$ the posterior mean converges to the MLE.
- Q3.** What is the difference between a 95% frequentist CI and a 95% Bayesian credible interval? Which allows a direct probability statement about θ ?

Quick Reference — Key Formulas

Probability

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A | B) = P(A \cap B) / P(B)$
- Independent: $P(A \cap B) = P(A)P(B)$
- Neither: $1 - P(A \cup B)$
- Bayes: $P(B_i | A) = \frac{P(A|B_i)P(B_i)}{\sum P(A|B_j)P(B_j)}$
- $\binom{n}{r} = n! / [r!(n-r)!]$

Descriptive Statistics

- $\bar{x} = \sum f_i m_i / \sum f_i$
- $s^2 = \sum f_i (m_i - \bar{x})^2 / (\sum f_i - 1)$
- $CV = (s / \bar{x}) \times 100\%$
- $M_r = L + \frac{h}{f} \left(\frac{rN}{k} - F \right)$

Estimation

- $\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$ (σ known)
- $\bar{x} \pm t_{\alpha/2, n-1} s / \sqrt{n}$ (σ unknown)
- χ^2 CI for σ^2 : $\left(\frac{(n-1)s^2}{\chi_{\alpha/2}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2} \right)$
- CRLB: $\text{Var}(\hat{\theta}) \geq 1/[nI(\theta)]$
- MLE: solve $\partial \ell / \partial \theta = 0$

Hypothesis Tests

- z : $(\bar{X} - \mu_0) / (\sigma / \sqrt{n})$
- t : $(\bar{X} - \mu_0) / (s / \sqrt{n}) \sim t(n-1)$
- Paired t : $\bar{D} / (s_D / \sqrt{n}) \sim t(n-1)$
- χ^2 : $(n-1)s^2 / \sigma_0^2 \sim \chi^2(n-1)$
- F : $s_1^2 / s_2^2 \sim F(n_1-1, n_2-1)$
- GOF: $\sum (O_i - E_i)^2 / E_i \sim \chi^2(k-1)$

Regression

- $\hat{\beta}_1 = S_{XY} / S_{XX}$
- $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$
- $r = S_{XY} / \sqrt{S_{XX} S_{YY}}$
- $R^2 = SSR / SST$
- r -test: $T = r \sqrt{n-2} / \sqrt{1-r^2} \sim t(n-2)$

Bayesian

- $\pi(\theta | x) \propto f(x | \theta) \pi(\theta)$
- Bayes est. (sq. loss) = $E(\theta | x)$
- Beta posterior: $\text{Beta}(\alpha + x, \beta + n - x)$
- Normal posterior: precision adds

Conf.	$z_{\alpha/2}$	z_{α}
90%	1.645	1.282
95%	1.960	1.645
99%	2.576	2.326

df	$t_{0.025}$	$t_{0.005}$
10	2.228	3.169
20	2.086	2.845
30	2.042	2.750
∞	1.960	2.576